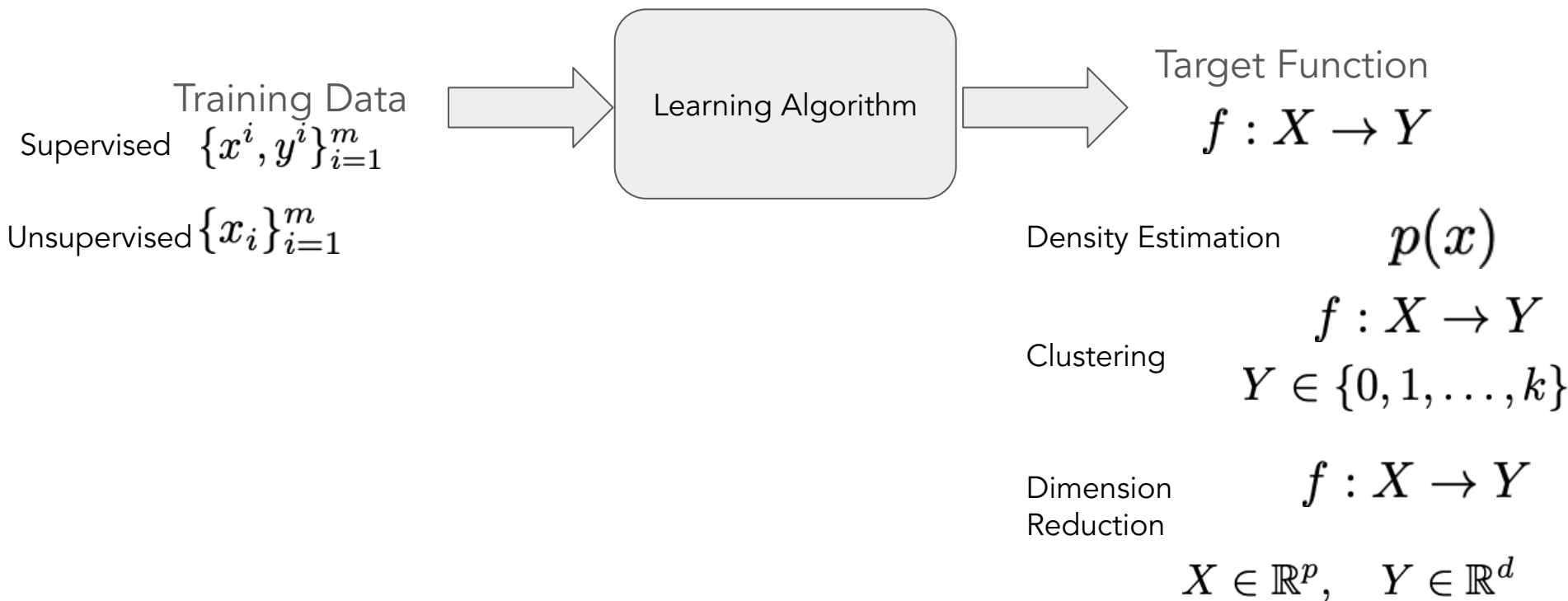


CX4240 Spring 2026

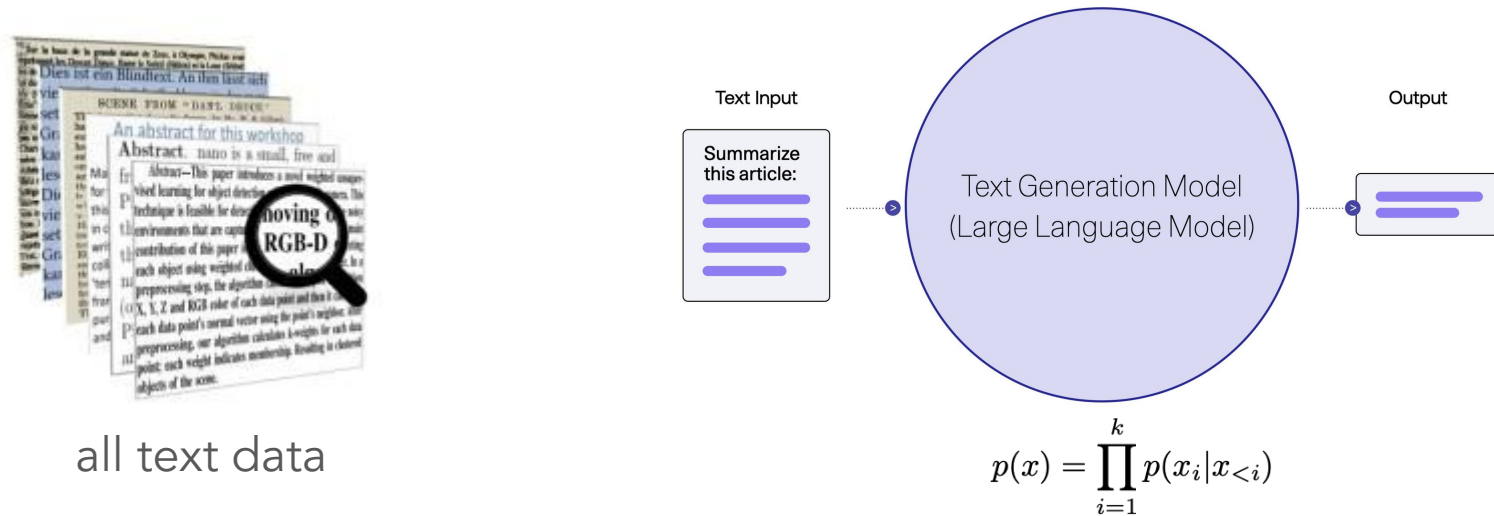
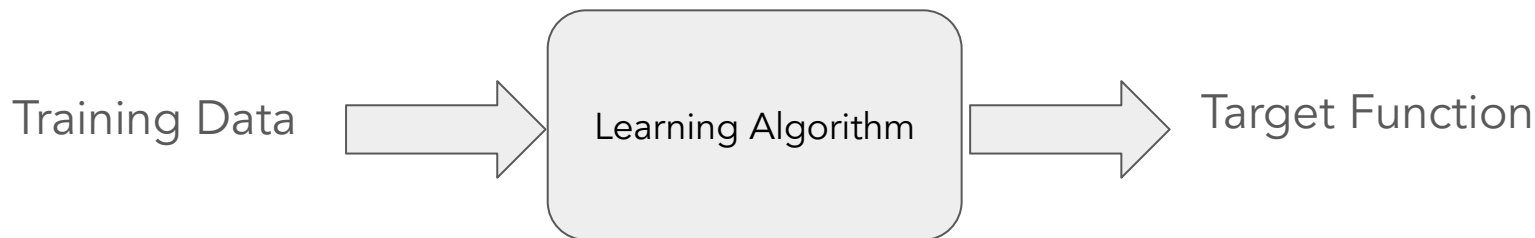
Generative LLM (Part I): Attention and Transformers

Bo Dai
School of CSE, Georgia Tech
bodai@cc.gatech.edu

Supervised Learning vs. Unsupervised Learning

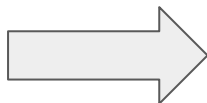


(Large) Language Models

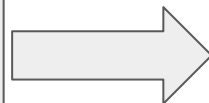


(Large) Language Models

Training Data



Learning Algorithm



Target Function



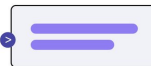
Text Input

Summarize
this article:



Text Generation Model
(Large Language Model)

Output



Unlabelled Data: text sequences

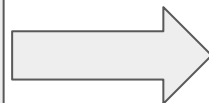
$$p(x) = \prod_{i=1}^k p(x_i | x_{<i})$$

(Large) Language Models

Training Data



Learning Algorithm



Target Function

Outcome (150-175 words):

Talk about your changing view on this belief or idea after the event you discussed in the Catalyst section (this is the 'after' picture you discussed in the Context section). If your old belief/idea was replaced with a new belief/idea, briefly explain what the new one is here.

Of course, avoidance doesn't fix anything. I had to face the music eventually and I have with time become more financially competent, spending hours as I can. I've started an education savings plan and a long better at spending my money; I no longer blow through a pay

I will spend my pay and exactly how much is allocated to ex know how to do yet. I know as I get older there will be financ can't even anticipate. I want to become better at budgeting want to diversify. I don't really know what 'diversify' means, i podcasts about finances and I read books. And when I have money.

Reflection (175-200 words):

Talk about what you learned by challenging this belief/opinion and forward. Provide some learning outcomes about how it impacted professionally.

Every adult has to deal with money. I felt so grown up when a paycheck with my name on it. However, the expectations believe it was a major shortcoming in my education to overt the hard way, slowly and with a lot of mistakes, how to man could have been better prepared prior if more adults around money. I started out believing money was something about but I've been slowly shedding those practices. I still find it di people, and when I do I still feel like I'm breaking a taboo, bi myself to speak plainly. And I have a goal: not only am I goi going to pay what I learn forward. When my kids earn their f know what to do with it. I know it's important, so I talk about

MATH Dataset (Ours)

Problem: Tom has a red marble, a green marble, a blue marble, and three identical yellow marbles. How many different groups of two marbles can Tom choose?

Solution: There are two cases here: either Tom chooses two yellow marbles (1 result), or he chooses two marbles of different colors ($\binom{4}{2} = 6$ results). The total number of distinct pairs of marbles Tom can choose is $1 + 6 = 7$.

Problem: If $\sum_{n=0}^{\infty} \cos^{2n} \theta = 5$, what is $\cos 2\theta$?

Solution: This geometric series is

$1 + \cos^2 \theta + \cos^4 \theta + \dots = \frac{1}{1 - \cos^2 \theta} = 5$. Hence,

$\cos^2 \theta = \frac{4}{5}$. Then $\cos 2\theta = 2 \cos^2 \theta - 1 = \frac{3}{5}$.

Problem: The equation $x^2 + 2x = i$ has two complex solutions. Determine the product of their real parts.

Solution: Complete the square by adding 1 to each side. Then $(x + 1)^2 = 1 + i = e^{\frac{\pi}{4}} \sqrt{2}$, so $x + 1 = \pm e^{\frac{\pi}{8}} \sqrt{2}$.

The desired product is then

$(-1 + \cos(\frac{\pi}{8}) \sqrt{2})(-1 - \cos(\frac{\pi}{8}) \sqrt{2}) =$

$1 - \cos^2(\frac{\pi}{8}) \sqrt{2} = 1 - \frac{(1 + \cos(\frac{\pi}{4}))}{2} \sqrt{2} = \frac{1 - \sqrt{2}}{2}$.

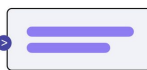
Text Input

Summarize this article:



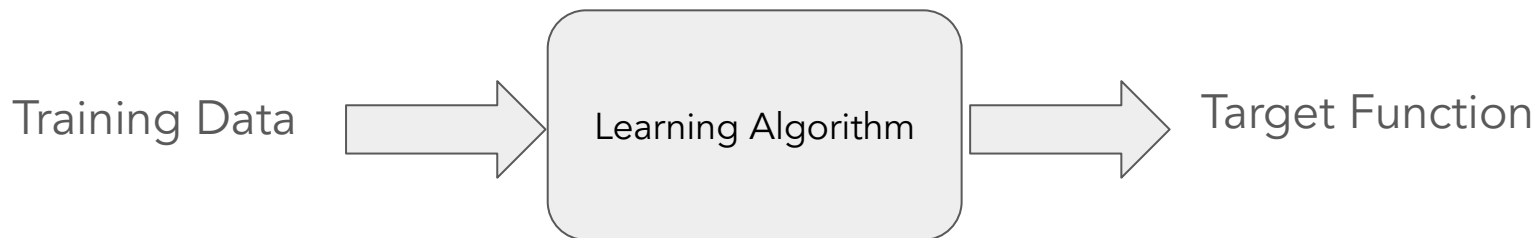
Text Generation Model
(Large Language Model)

Output



Labelled Data: Prompt & Answer

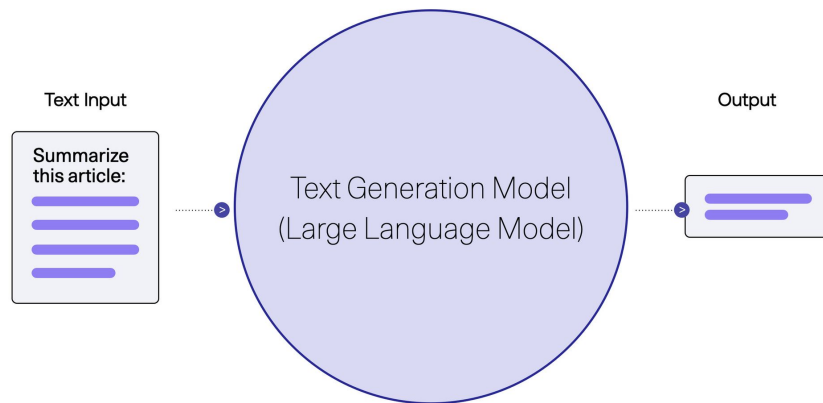
(Large) Language Models



What are the key benefits of using Reinforcement Learning from Human Feedback (RLHF) for dataset collection in the context of Large Language Model (LLM) generation?

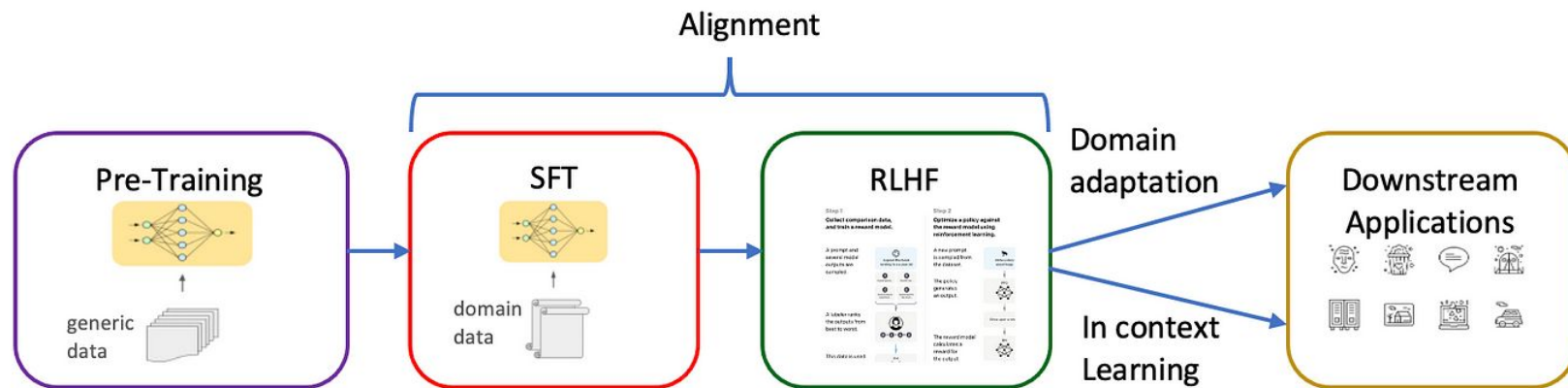
Reinforcement Learning from Human Feedback (RLHF) for dataset collection in Large Language Model (LLM) generation provides key benefits such as improved model performance through direct optimization, better alignment with human values by incorporating human feedback, and the ability to iteratively refine the model based on user interactions, resulting in a more user-friendly and efficient language model.

Using reinforcement Learning from Human Feedback (RLHF) for dataset collection in Large Language Model (LLM) generation offers advantages such as enhanced model capabilities by optimizing for desired outcomes, greater adaptability to human preferences through the inclusion of human feedback, and the opportunity to continuously improve the model based on user experiences, ultimately leading to a more effective and responsive language model.

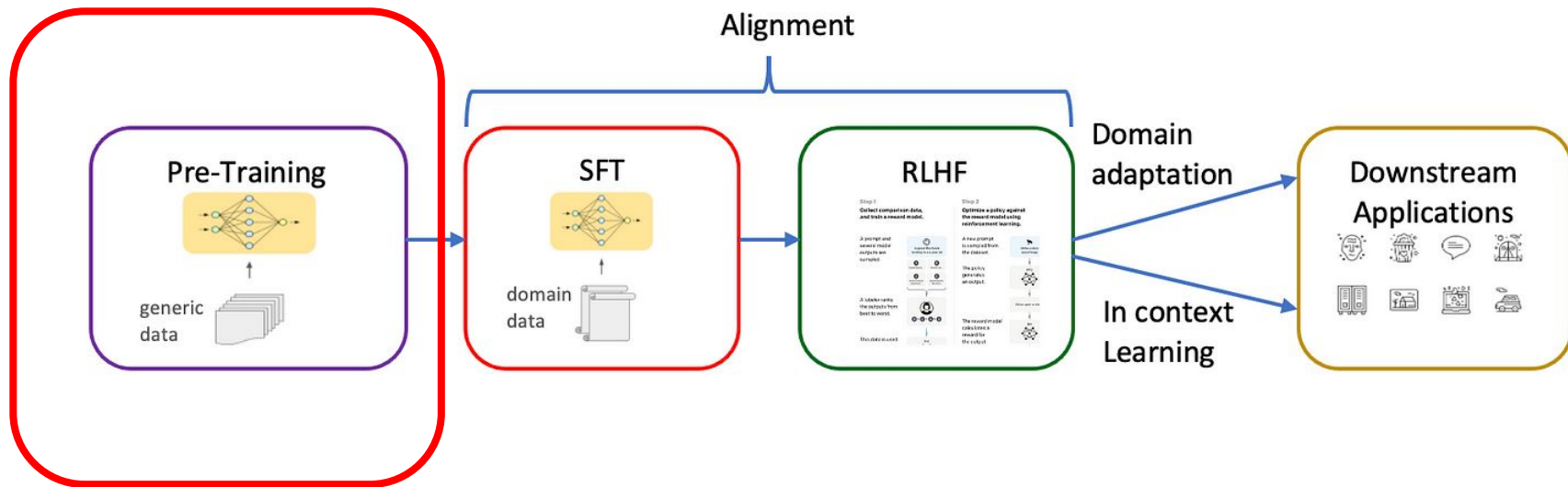


Preference Data: Prompt & Positive/ Negative Answers

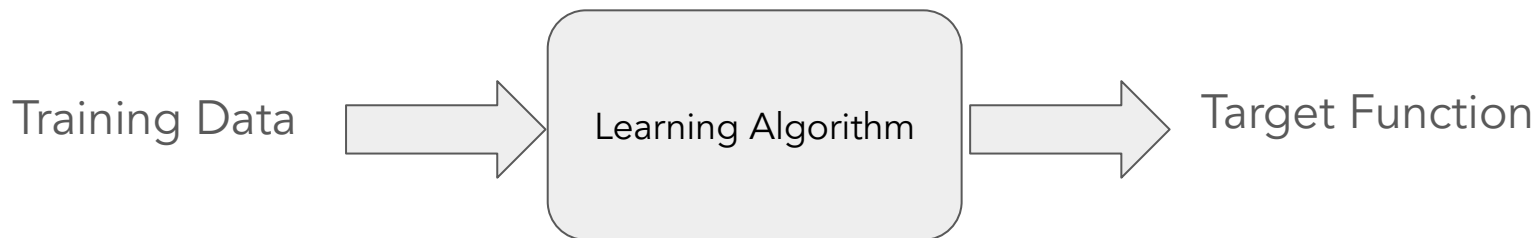
LLM Training



LLM Training



(Large) Language Models

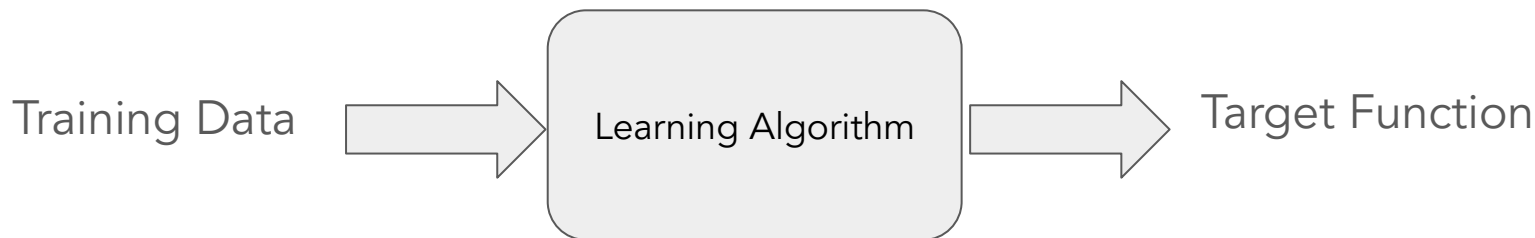


For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

$p(X)$

$$D = \{X^i\}_{i=1}^n, \quad X^i = \{x_j^i\}_{j=1}^t$$

(Large) Language Models



$$D = \{X^i\}_{i=1}^n, \quad X^i = \{x_j^i\}_{j=1}^t$$

$$p(X)$$

1. Build probabilistic models
Categorical Distribution + Autoregressive +
RNN/Transformer
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

Combating Combinatorial Complexity: Autoregressive Model

$$p(X) = p(\{x_j\}_{j=1}^t) \quad O(|V|^T)$$

Combating Combinatorial Complexity: Autoregressive Model

$$\begin{aligned} p(X) &= p(\{x_j\}_{j=1}^t) && O(|V|^T) \\ &= \prod_{j=1}^t p(x_j | x_{<j}) \end{aligned}$$

Categorical Distribution

$$\begin{aligned} p(X) &= p(\{x_j\}_{j=1}^t) && O(|V|^T) \\ &= \prod_{j=1}^t p(x_j | x_{<j}) \\ &= \prod_{j=1}^t \frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))} \end{aligned}$$

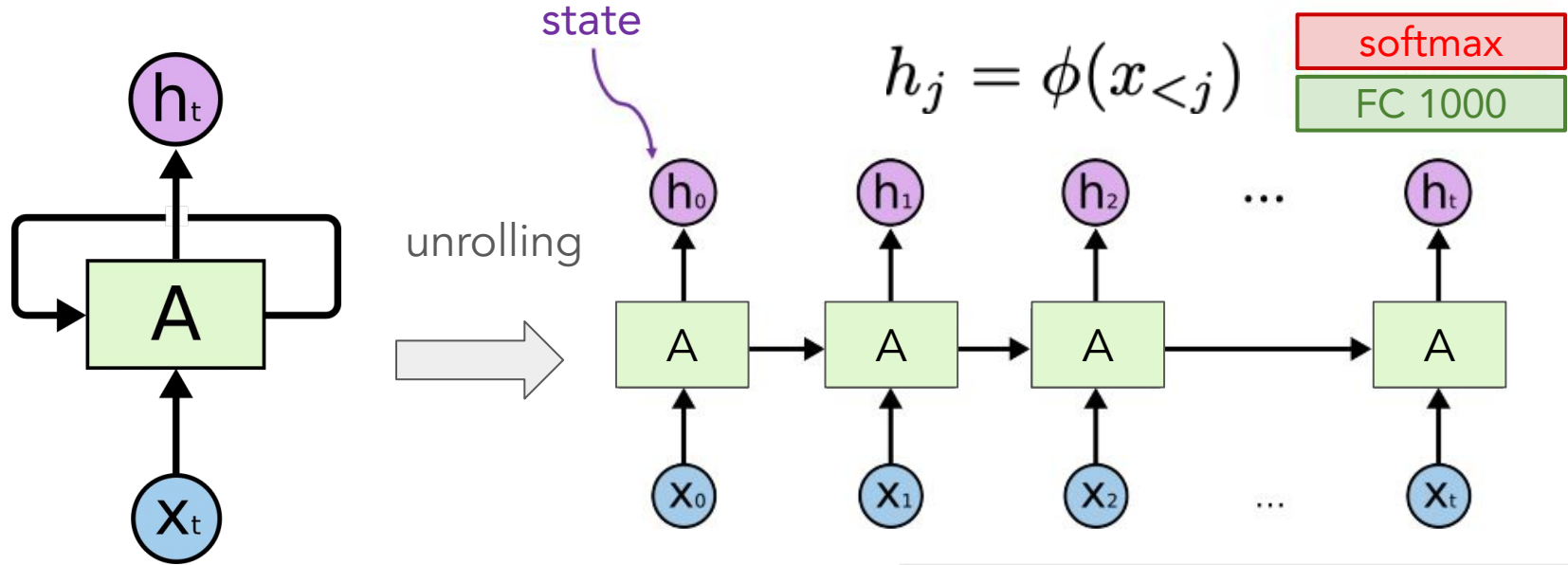
Recursive Neural Network

$$\begin{aligned} p(X) &= p(\{x_j\}_{j=1}^t) && O(|V|^T) \\ &= \prod_{j=1}^t p(x_j | x_{<j}) \\ &= \prod_{j=1}^t \frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))} \end{aligned}$$

RNN

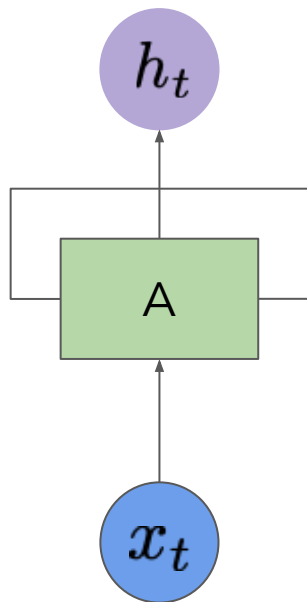
Recurrent Neural Network

$$\frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))}$$

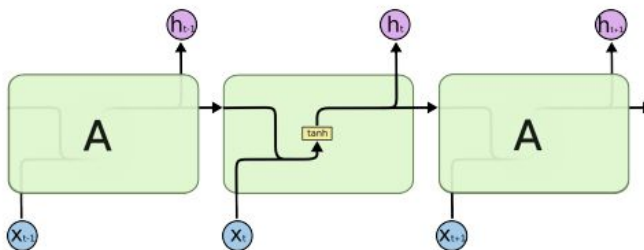


For a movie that gets no respect there sure are a lot of memorable quotes listed for this gem. Imagine a movie where Joe Piscopo is actually funny! Maureen Stapleton is a scene stealer. The Moroni character is an absolute scream. Watch for Alan "The Skipper" Hale jr. as a police Sgt.

RNN Cell



$$\frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))}$$

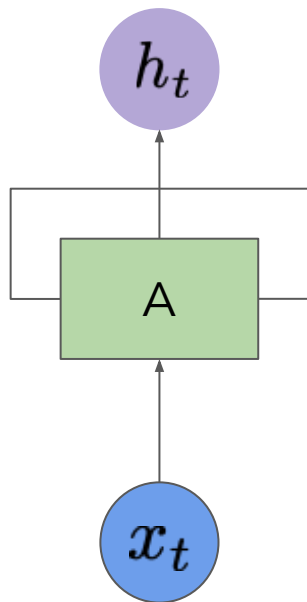


Simple RNN

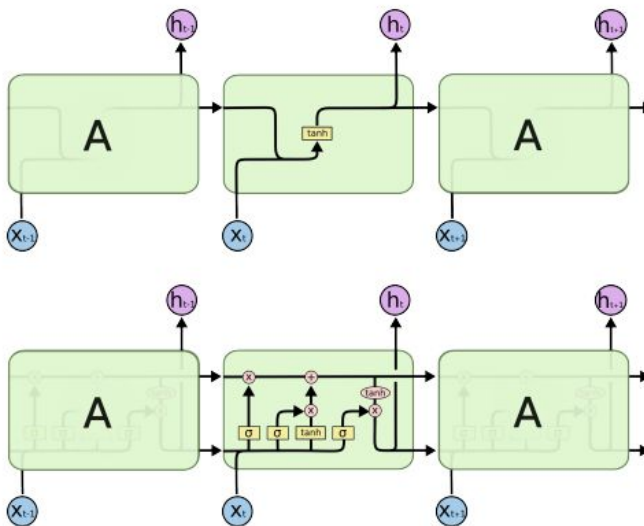
$\mathbf{h}_{100}$ is almost irrelevant to $\mathbf{x}_1$: $\frac{\partial \mathbf{h}_{100}}{\partial \mathbf{x}_1}$ is near zero.

Gradient Vanishing

RNN Cell



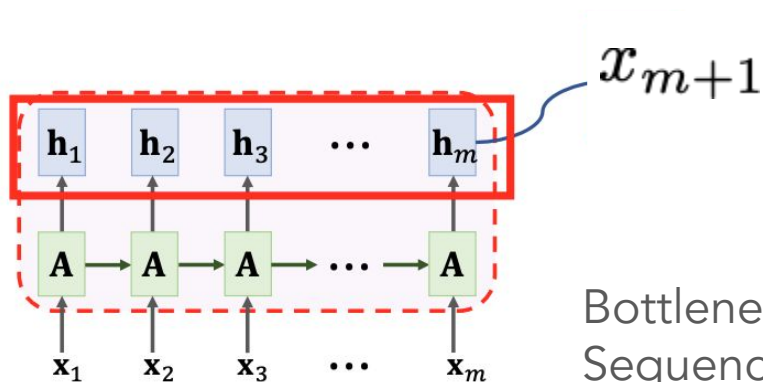
$$\frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))}$$



Simple RNN

LSTM

Bottleneck in RNN

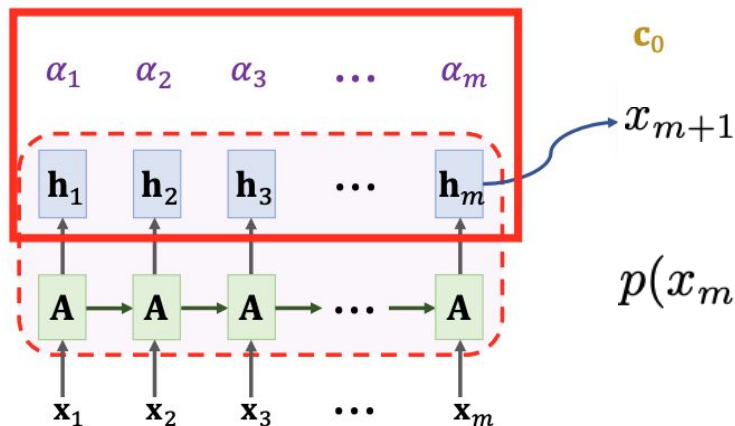


$$h_j = \phi(x_{<j})$$
$$\frac{\exp(W_{x_j} \phi(x_{<j}))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}))}$$

Bottleneck:
Sequences bottlenecked through a
fixed-sized vector.

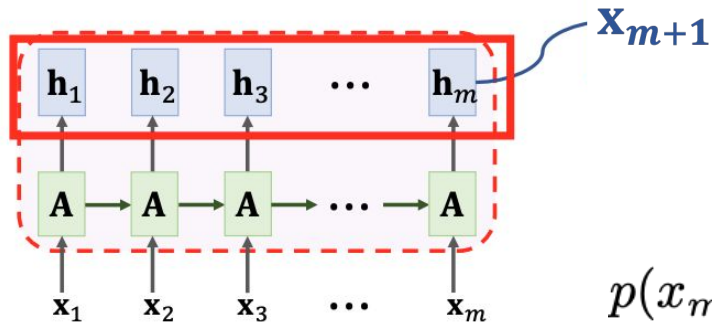
Attention: Flatten RNN Computation

Context vector: $\mathbf{c}_0 = \alpha_1 \mathbf{h}_1 + \dots + \alpha_m \mathbf{h}_m$.



$$p(x_{m+1} | x_{<m+1}) = \frac{\exp(W_{x_j} c_0)}{\sum_{l=1}^V \exp(W_l c_0)}$$

Attention: Adding Output Dependency

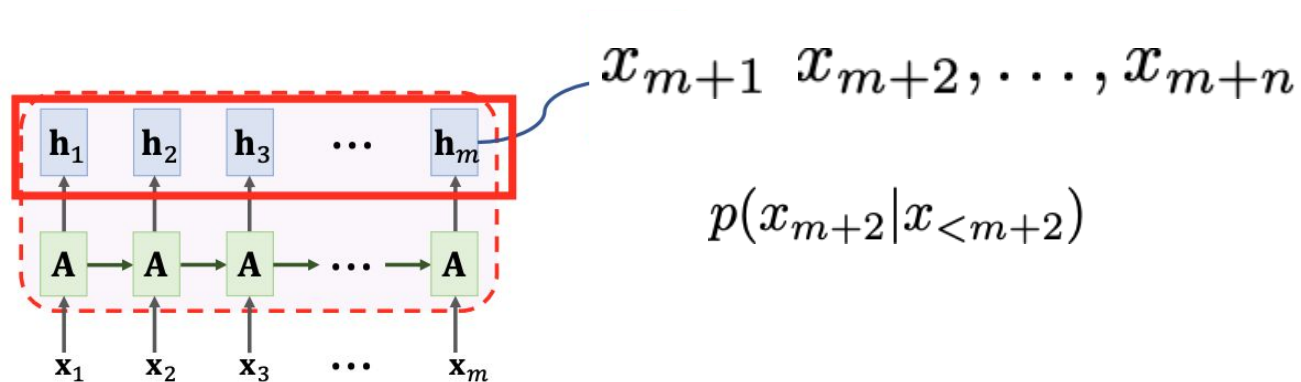


1. Linear maps:
 - $\mathbf{k}_i = \mathbf{W}_K \cdot \mathbf{h}_i$, for $i = 1$ to m .
 - $\mathbf{q}_0 = \mathbf{W}_Q \cdot \mathbf{x}_{m+1}$
2. Inner product:
 - $\tilde{\alpha}_i = \mathbf{k}_i^T \mathbf{q}_0$, for $i = 1$ to m .
3. Normalization:
 - $[\alpha_1, \dots, \alpha_m] = \text{Softmax}([\tilde{\alpha}_1, \dots, \tilde{\alpha}_m])$.

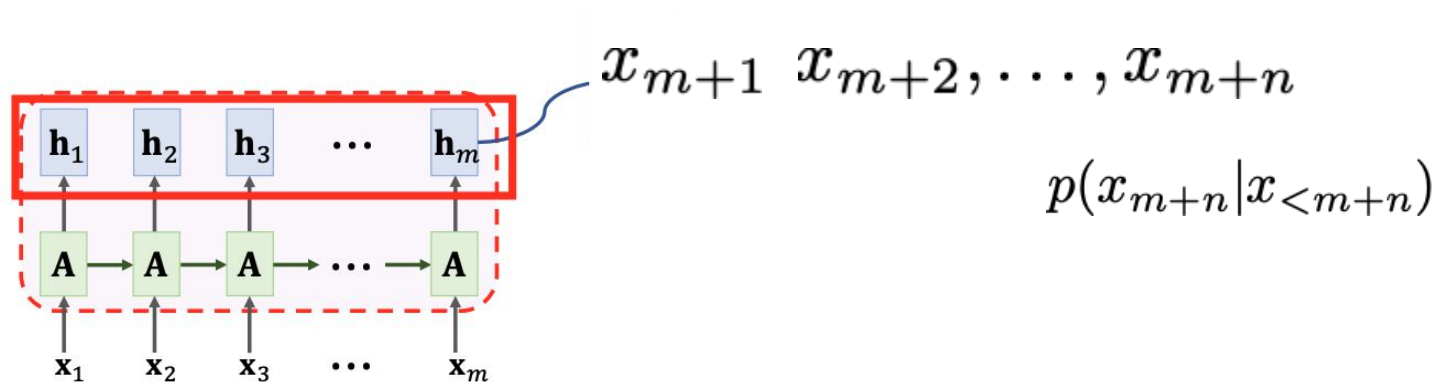
$$p(x_{m+1} | x_{<m+1}) = \frac{\exp(W_{x_j} c_0)}{\sum_{l=1}^V \exp(W_l c_0)}$$

Weight: $\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{x}_{m+1})$

Recursive Attention?



Recursive Attention?

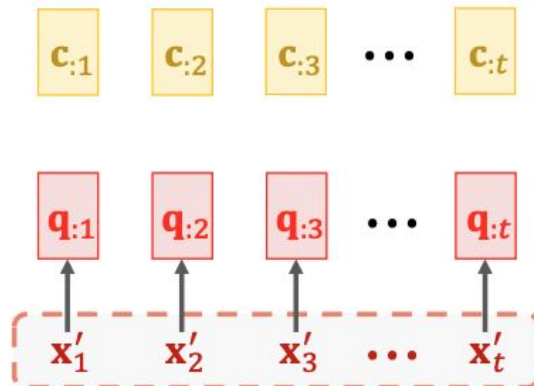
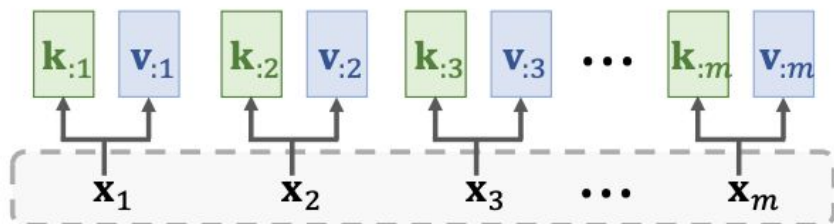


Parallel Attention

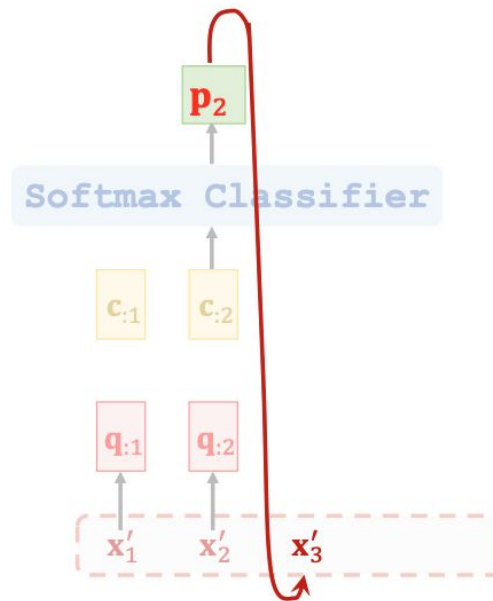
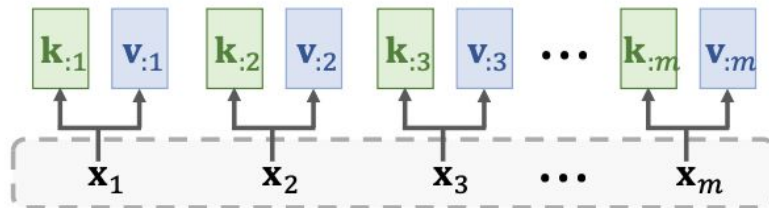
Query: $\mathbf{q}_{:j} = \mathbf{W}_Q \mathbf{x}_j$ **Key:** $\mathbf{k}_{:i} = \mathbf{W}_K \mathbf{h}_i$, **Value:** $\mathbf{v}_{:i} = \mathbf{W}_V \mathbf{h}_i$.

Weights: $\alpha_{:j} = \text{Softmax}(\mathbf{K}^T \mathbf{q}_{:j}) \in \mathbb{R}^m$.

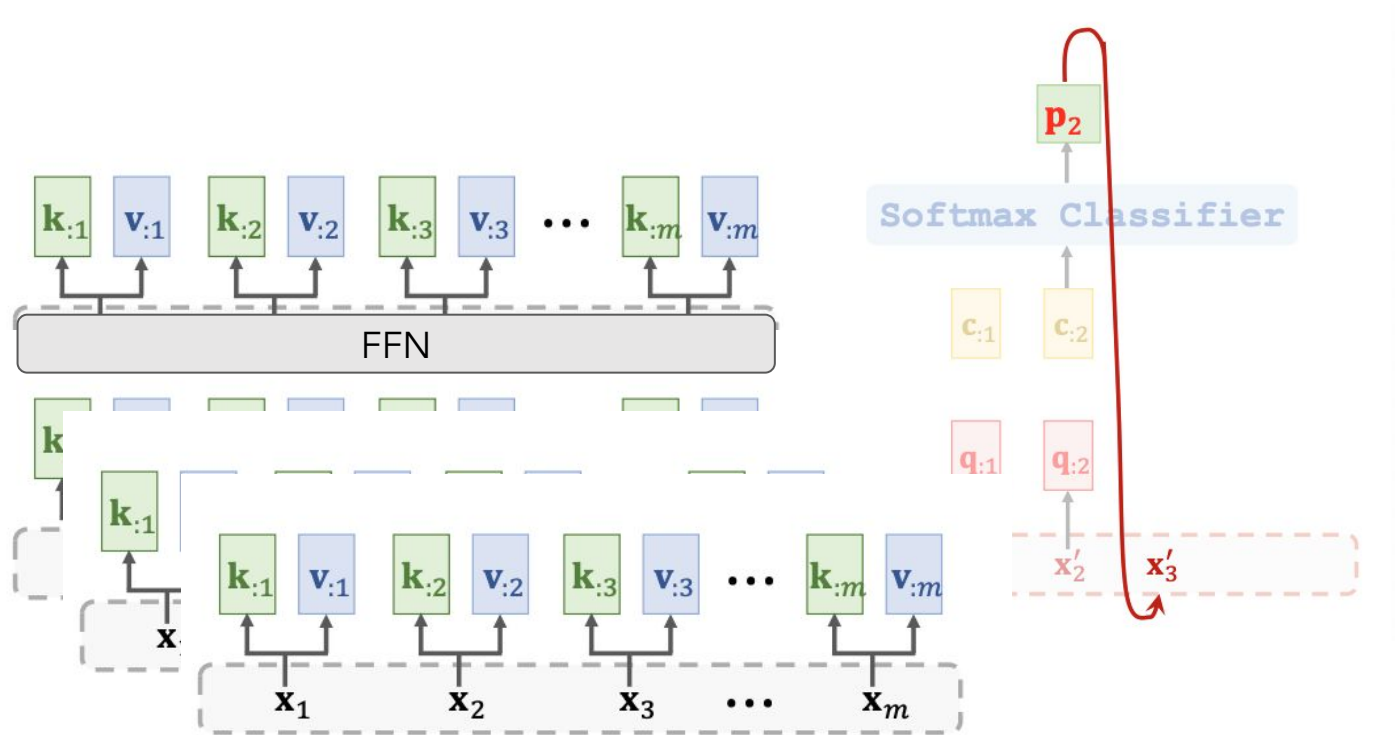
Context vector: $\mathbf{c}_j = \alpha_{1j} \mathbf{v}_{:1} + \dots + \alpha_{mj} \mathbf{v}_{:m}$.



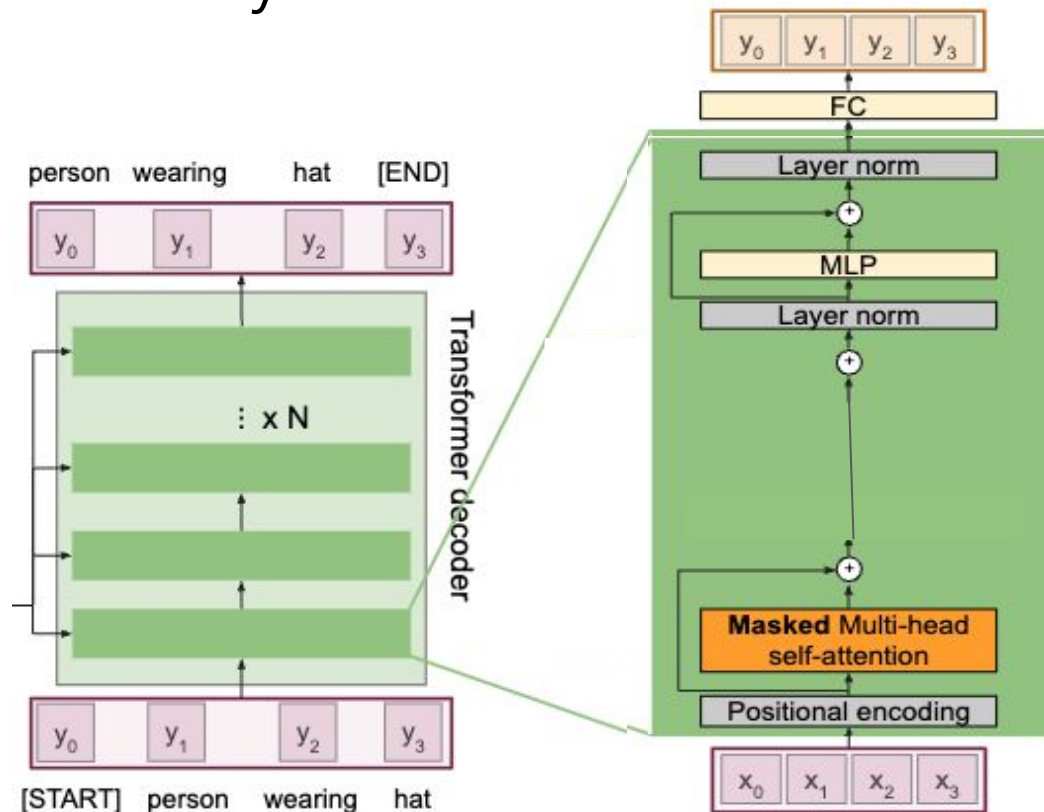
Parallel Attention



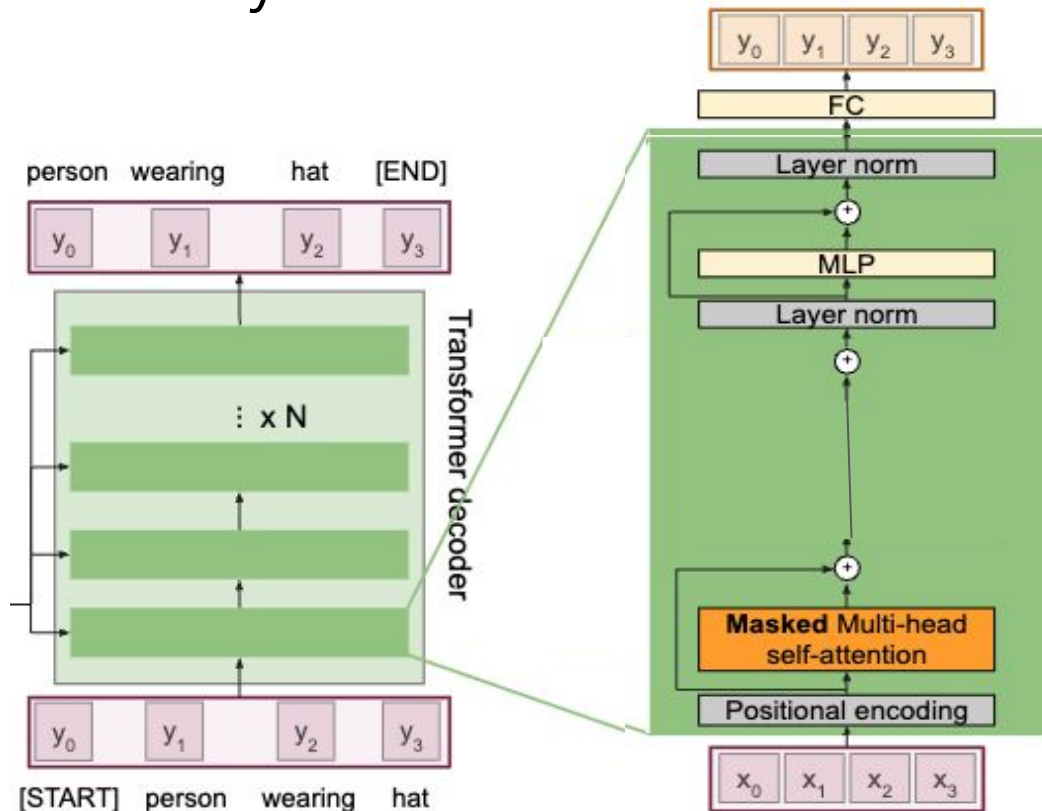
Transformer: Multi-Headed Multi-Layer Parallel Attention!



Decoder-Only Transformer



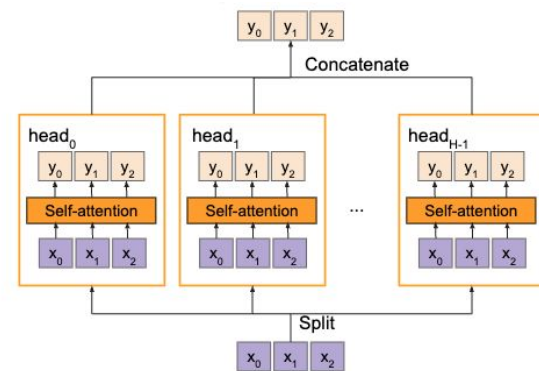
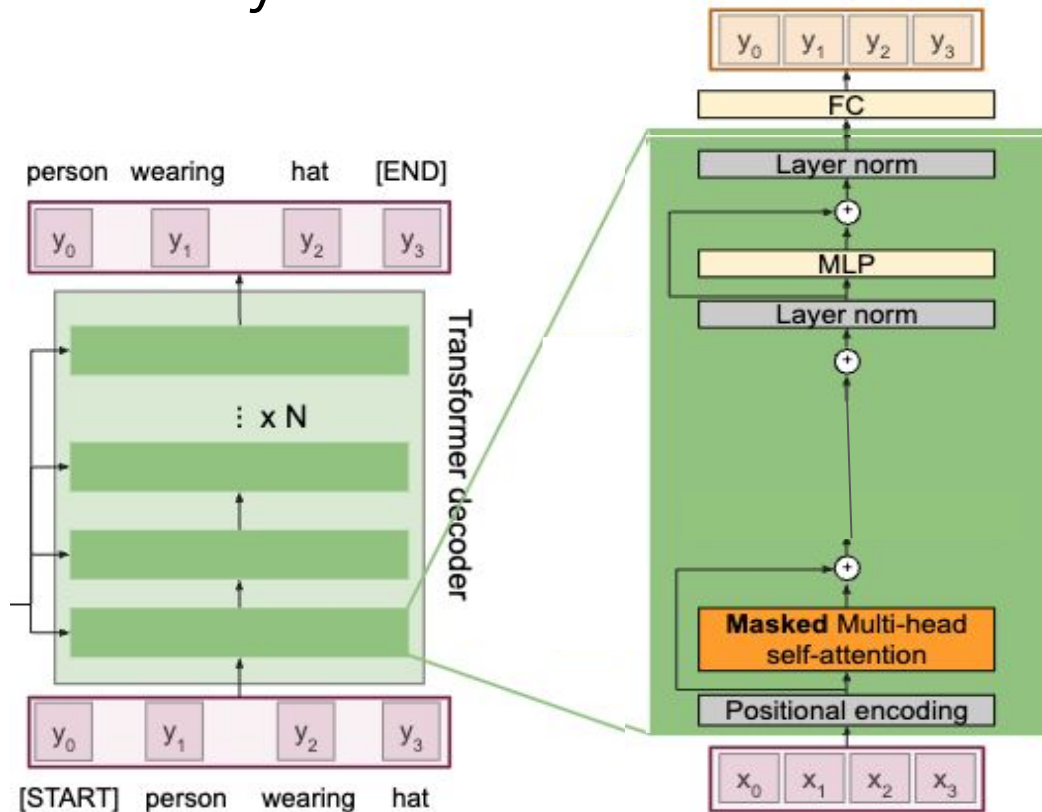
Decoder-Only Transformer



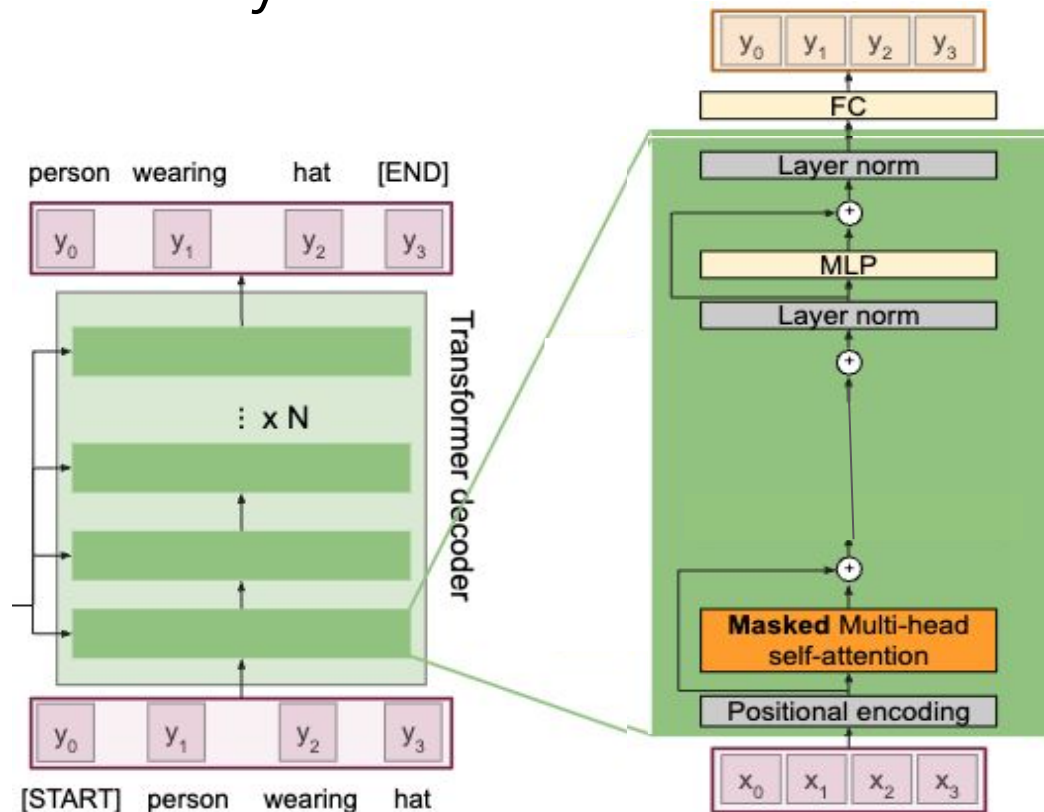
$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Decoder-Only Transformer



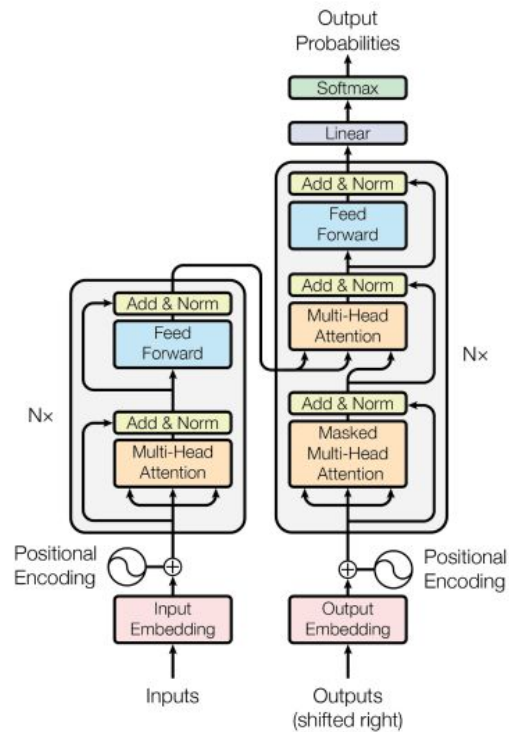
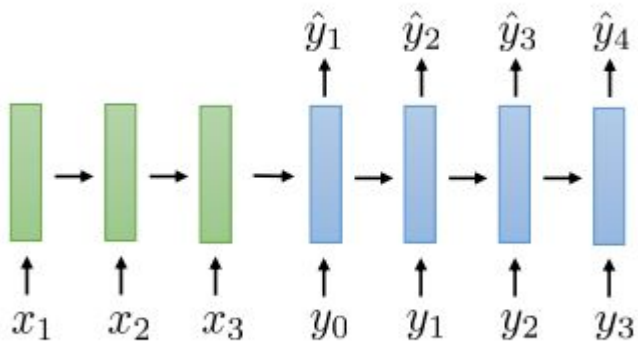
Decoder-Only Transformer



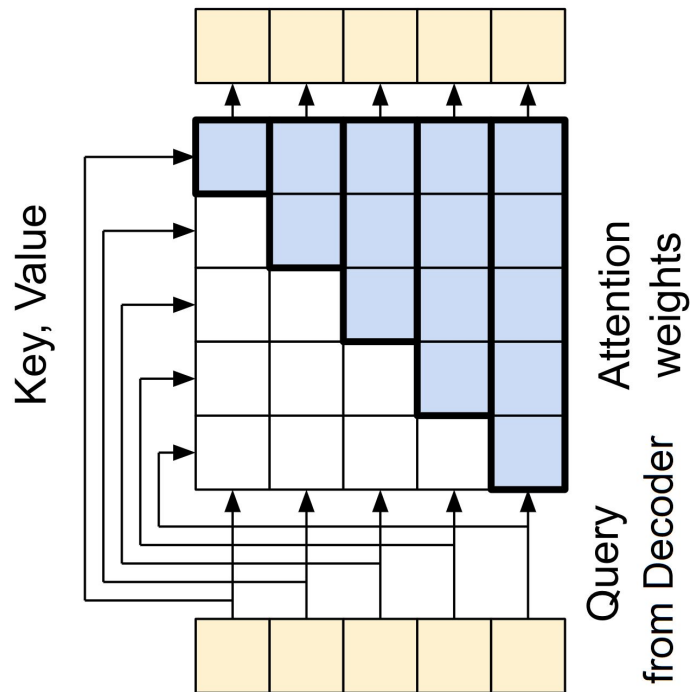
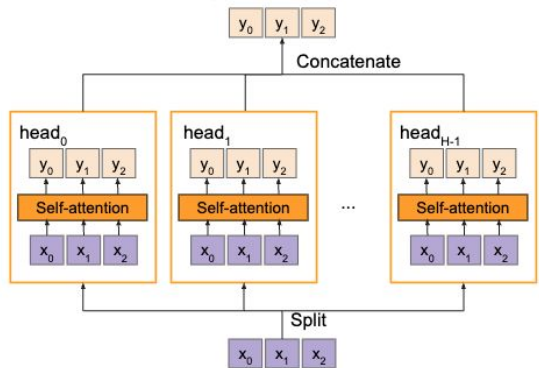
residual connect

residual connect

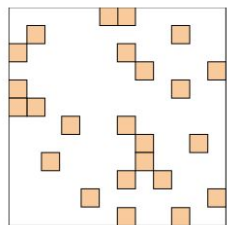
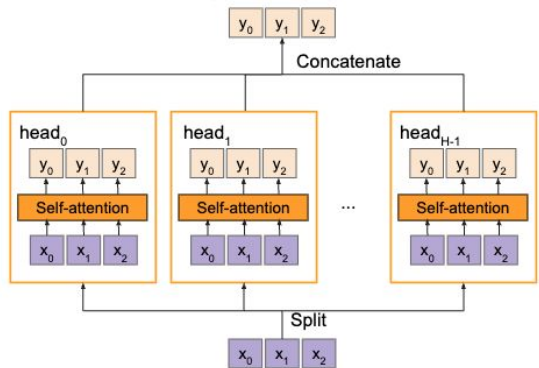
Original Transformer



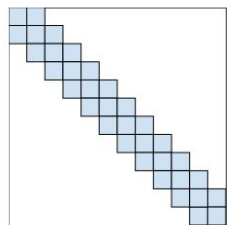
Limitations? Expensive Computation



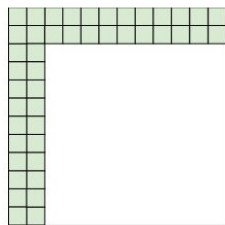
Limitations?



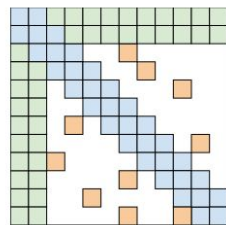
(a) Random attention



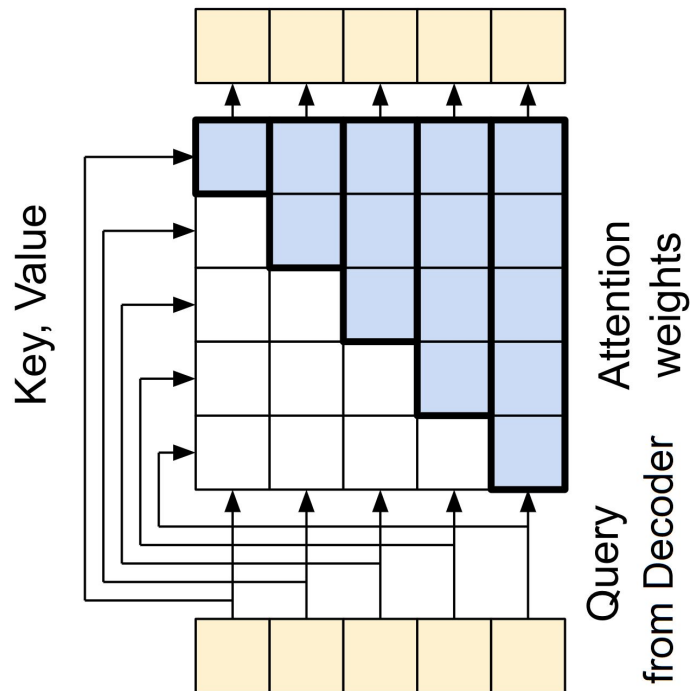
(b) Window attention



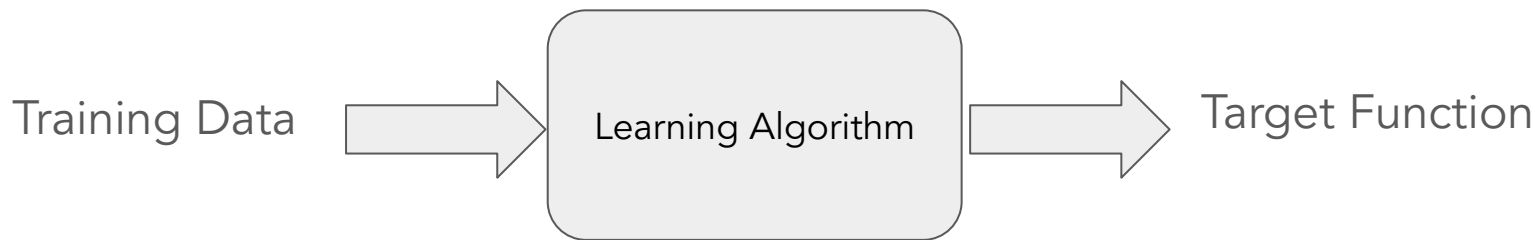
(c) Global Attention



(d) BIGBIRD



(Large) Language Models



$$D = \{X^i\}_{i=1}^n, \quad X^i = \{x_j^i\}_{j=1}^t$$

$$p(X)$$

1. Build probabilistic models
2. Derive loss function (by MLE or MAP....)
MLE
3. Select optimizer

MLE

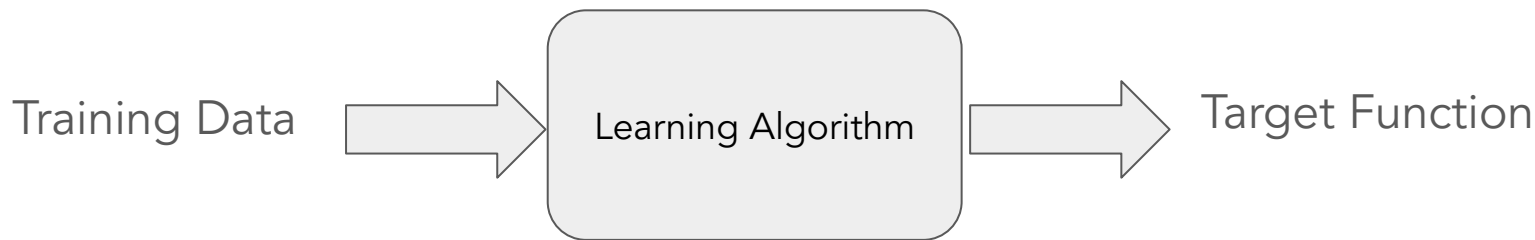
- Given all input data $D = \{X^i\}_{i=1}^n$, $X^i = \{x_j^i\}_{j=1}^t$

$$p(X^i) = \prod_{j=1}^t \frac{\exp(W_{x_j^i} \phi(x_{<j}^i))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}^i))}$$

- Log-likelihood

$$\begin{aligned} \ell(\phi) &= \sum_{i=1}^n p(X^i; \phi) = \sum_{i=1}^n \sum_{j=1}^t \log(x_j^i \mid x_{<j}^i; \phi) \\ &= \sum_{i=1}^n \sum_{j=1}^t \log \frac{\exp(W_{x_j^i} \phi(x_{<j}^i))}{\sum_{l=1}^V \exp(W_l \phi(x_{<j}^i))} \\ &= \sum_{i=1}^n \sum_{j=1}^t W_{x_j^i} \phi(x_{<j}^i) - \sum_{i=1}^n \sum_{j=1}^t \log \sum_{l=1}^V \exp(W_l \phi(x_{<j}^i)) \end{aligned}$$

(Large) Language Models



$$D = \{X^i\}_{i=1}^n, \quad X^i = \{x_j^i\}_{j=1}^t$$

$$p(X)$$

1. Build probabilistic models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

Stochastic Gradient Descent

Summary

- Pretraining of LLM: MLE for unsupervised learning
- RNNs requires sequential processing
- RNNs have bottleneck, restricting the flexibility
- Attention is actually flattening RNN
- Transformer is deep attention

Summary

- Pretraining of LLM: MLE for unsupervised learning
- RNNs requires sequential processing
- RNNs have bottleneck, restricting the flexibility
- Attention is actually flattening RNN
- Transformer is deep attention
- There are **other LLMs**: Encoder-Decoder, Encoder-only, etc...

Q&A

Presentation

Sign-up deadline
today!